

AI Lab Assignment 2

Submission Guidelines-

- Rename the file to your ID only. If your ID is 18-XXXXX-1, then the file name must be 18-XXXXX-1.pdf.
- Must submit within the time in Teams.
- Must include resources for all the sections in the table and Uploaded file should be in PDF format.

Question- 1

Perform **BFS** algorithms for the given graph where start node is A and Goal node is K.
Graph with Weights:

A: [(B, 2), (C, 1)]

B: [(A, 2), (D, 4), (E, 6)]

C: [(A, 1), (F, 5), (G, 2)]

D: [(B, 4), (H, 3)]

E: [(B, 6), (I, 2)]

F: [(C, 5)]

G: [(C, 2), (J, 7)]

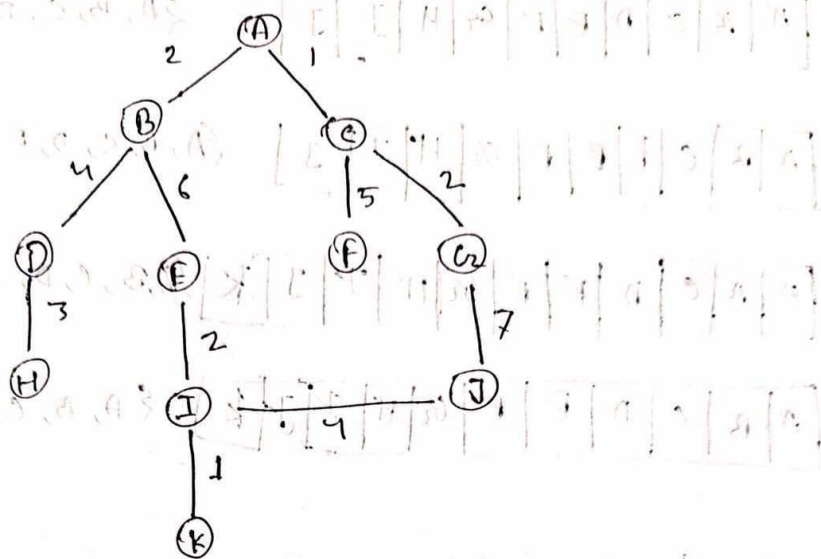
H: [(D, 3)]

I: [(E, 2), (J, 4), (K, 1)]

J: [(G, 7), (I, 4)]

K: [(I, 1)]

Solution (Do it on paper, and the solution should be attached here. For the theory simulation , you have to draw the graph and find frontier, explored set, path(solution), Path Cost, Tree):



Frontier

A

A B

A B C D E

A B C D E F G

A B C D E F G H

A B C D E F G H I

A B C D E F G H I

Explored state

$\{A\}$

$\{A, B\}$

$\{A, B, C\}$

$\{A, B, C, D\}$

$\{A, B, C, D, E\}$

$\{A, B, C, D, E, F\}$

A	B	C	D	E	F	G	H	I	J
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$\{A, B, C, D, E, F, G\}$

A	B	C	D	E	F	G	H	I	J
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$\{A, B, C, D, E, F, G, H\}$

A	B	C	D	E	F	G	H	I	J	K
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$\{A, B, C, D, E, F, G, H, I\}$

A	B	C	D	E	F	G	H	I	J	K
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$\{A, B, C, D, E, F, G, H, I, J\}$

$\therefore \text{Total Path cost} = A + B +$

$\therefore \text{Path } A \rightarrow D \rightarrow E \rightarrow I \rightarrow K$

Path cost = $A + B + E + I + K$

$$= 2 + 2 + 6 + 2 + 1$$

$$= 11$$

Code (For coding implementation, show the total path cost and path from A to K.)-

```
from collections import deque
```

```
graph = {  
    'A': [('B', 2), ('C', 1)],  
    'B': [('A', 2), ('D', 4), ('E', 6)],  
    'C': [('A', 1), ('F', 5), ('G', 2)],  
    'D': [('B', 4), ('H', 3)],  
    'E': [('B', 6), ('I', 2)],  
    'F': [('C', 5)],  
    'G': [('C', 2), ('J', 7)],  
    'H': [('D', 3)],  
    'I': [('E', 2), ('J', 4), ('K', 1)],  
    'J': [('G', 7), ('I', 4)],  
    'K': [('I', 1)]  
}
```

```
def bfs_with_cost(graph, start, goal):  
    queue = deque([(start, [start])])  
    visited = set()
```

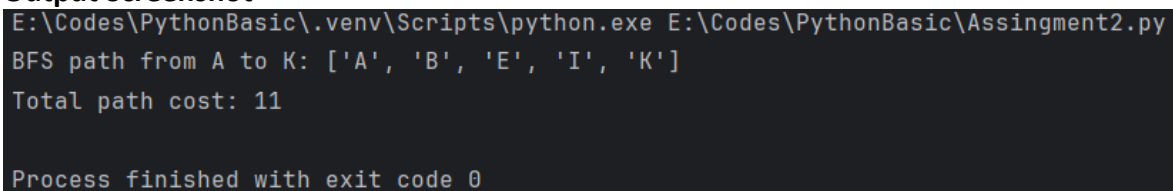
```
    while queue:  
        node, path = queue.popleft()  
        if node == goal:  
            return path, calculate_cost(graph, path)  
        if node not in visited:  
            visited.add(node)  
            for neighbor, _ in graph[node]:  
                if neighbor not in visited and neighbor not in path:  
                    queue.append((neighbor, path + [neighbor]))  
    return None, 0
```

```
def calculate_cost(graph, path):  
    cost = 0  
    for i in range(len(path) - 1):  
        current = path[i]  
        next_node = path[i + 1]  
        for neighbor, weight in graph[current]:  
            if neighbor == next_node:  
                cost += weight
```

```
        break
    return cost
```

```
# Run BFS from A to K
path, total_cost = bfs_with_cost(graph, 'A', 'K')
print("BFS path from A to K:", path)
print("Total path cost:", total_cost)
```

Output Screenshot-



```
E:\Codes\PythonBasic\.venv\Scripts\python.exe E:\Codes\PythonBasic\Assingment2.py
BFS path from A to K: ['A', 'B', 'E', 'I', 'K']
Total path cost: 11

Process finished with exit code 0
```



